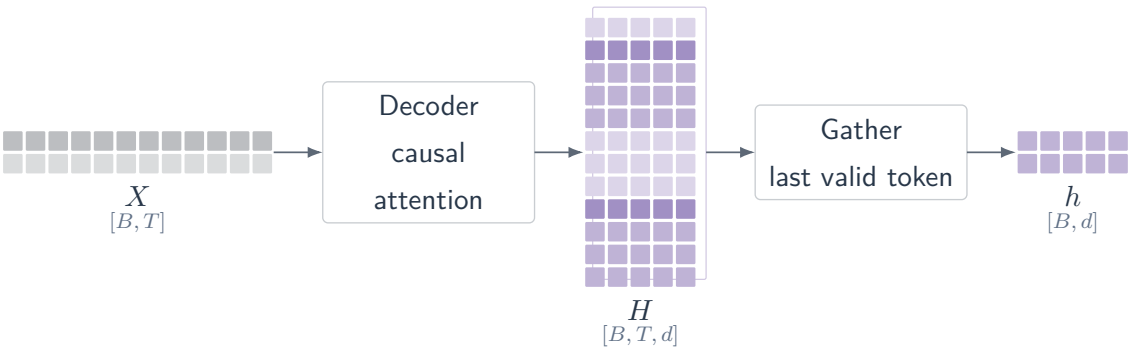
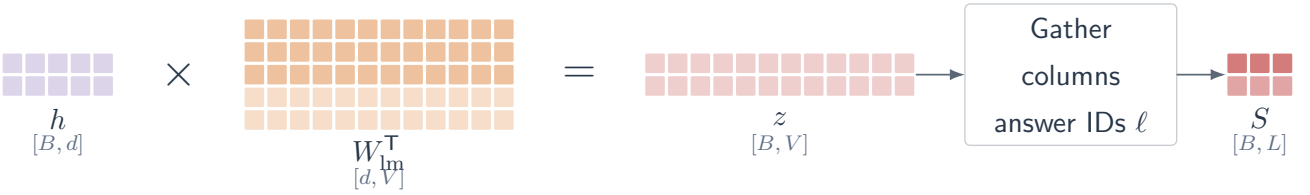
$$h_b = H_{b,t_b^\star,:}, \quad z = hW_{\text{lm}}^{\mathsf{T}}, \quad S_{b,k} = z_{b,\ell_{b,k}}, \quad p_b = \text{softmax}_k(S_b/\tau)$$

Causal encoding and last-position selection



Vocabulary projection, then ordered answer-token selection



Axes	B: questions; T: tokens; d: hidden channels; V: vocabulary; L: answer-token IDs. W_{lm} is stored as $[V, d]$.
Meaning	The readout position sees all preceding options. Choice and score normalize over selected answer tokens, not the full vocabulary. For noul, the implementation uses $z_{yes} - z_{no}$ followed by sigmoid.

Cell counts illustrate axes, not model sizes; tint identifies roles, not numeric values.